

A Native Behavioral Basis for the Assistant Persona

LLM-native factors explain and steer the Assistant Axis far better than the Big Five

Cues-Behavior project • meta-llama/Llama-3.3-70B-Instruct

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Abstract

Our earlier work found that the five-factor (Big Five / OCEAN) model explains only $\approx 29\%$ of where a persona sits on the *Assistant Axis* (AA) — the dominant direction in Llama-3.3-70B’s persona space that indexes how strongly the model is in its default helpful-Assistant mode. Motivated by the 2024+ literature arguing that Big Five is the wrong basis for the *assistant* persona, we test an alternative: a small set of **LLM-native behavioral directions** (verbosity, deference, boldness, guardedness, warmth, groundedness, formality), built persona-vector / contrastive-activation style. Across the 275 personas of the Assistant-Axis atlas, seven native factors explain $R^2 = 0.96$ of the Assistant Axis versus 0.28 for the Big Five; a single native factor, *groundedness* (literal/factual $\leftrightarrow$ mystical/dramatic), explains 0.77 on its own — more than all five personality traits combined. A causal test confirms the correlation: steering *groundedness* (at a layer band that excludes the readout) moves the Assistant Axis with a swing of +8.6, roughly $2.7\times$ the strongest Big Five direction built and steered identically, is monotone in dose, and behaviorally converts the default Assistant into an **eldritch** dramatic character and back. The assistant persona is a *behavioral-style* axis (grounded and forthcoming), not a personality-trait combination.

1 Motivation

The Assistant Axis (AA) is the top component of the model’s persona space: steering toward it reinforces helpful-harmless behavior, steering away makes the model adopt other identities and speaking styles [1]. Our H3 decomposition regressed each persona’s AA position on its five Big Five coordinates and found $R^2 = 0.29$ — a 71% “AI-ness” residual orthogonal to the human-personality map. The recent literature offers an explanation and an alternative: LLM “personality” self-reports do not predict behavior [4, 5], and the natural axes of model behavior are not OCEAN but *native behavioral factors* (Contreras’s Responsiveness / Deference / Boldness / Guardedness / Verbosity; 4), while “who the model is being” is increasingly modeled as an activation *direction* [2, 3]. This report asks whether a native behavioral basis *describes* and *controls* the Assistant Axis better than the Big Five.

2 Data

All measurements use meta-llama/Llama-3.3-70B-Instruct. The persona set is the **275-role Assistant-Axis atlas** (274 named character roles + the default Assistant); for each role we have the residual-stream response-mean activation (GEN-MEAN) at all 80 layers, shape $[80 \times 8192]$, from role-play rollouts. The Assistant Axis $\hat{a}$ is the unit axis vector at layer 40 (`assistant_axis.npy`); the Big Five probes are per-layer M2 ridge directions at their probe-optimal layers (EXT@30,

AGR@31, CSN@31, EST@30, OPN@36). Neutral prompts for the causal stage are 16 held-out extraction questions posed to the default Assistant (no persona).

3 Method

Native behavioral directions. For each of seven factors we write a *high-pole* and *low-pole* system instruction (e.g. groundedness: “grounded, literal and factual” vs. “mystical, abstract and dramatic”), pose 12 shared neutral questions under each pole, read the response activation $\text{GEN-MEAN}[L]$, and take the mean difference $\text{dir} = \overline{\text{high}} - \overline{\text{low}}$ (unit-normalized). This is the Persona-Vectors / Contrastive-Activation-Addition recipe [2, 6]. The factors are deliberately behavioral *styles*, none of them “helpfulness”, to avoid a tautology with the Assistant Axis.

One consistent projection basis. Every one of the 275 role vectors is projected *from the same* $[80 \times 8192]$ vector onto (a) the Assistant Axis ($\text{role}[40] \cdot \hat{a}$), (b) the Big Five probes at their probe layers, and (c) the seven native directions at layer 40. We then regress the AA position (standardized) on each block across the 274 role vectors and report R^2 , incremental R^2 , and per-factor Pearson r .

Control for a layer/method confound. The native directions live at layer 40, the same layer as the AA readout, whereas the Big Five probes live at layers 30–36. To isolate whether any advantage is the *factor choice* or merely the layer/method, we rebuild the Big Five at layer 40 with the identical contrastive recipe (a “Big Five @L40” control) and regress AA on those directions too.

Causal test. To ask whether the correlation is causal we steer each factor during generation. A naïve steer at the readout layer would be mechanically trivial (adding a vector that overlaps $\hat{a}$ inflates $\text{role}[40] \cdot \hat{a}$ by construction). We therefore build *per-layer* direction banks and steer only the band [24,36], **which excludes the readout layer 40**, so any shift in AA is a genuinely *computed* downstream effect. On the 16 neutral prompts we steer each factor at $\pm c$ (norm-scaled, $c = 0.12$) and read the response’s AA projection, its nearest evoked role (cosine to the 275 role vectors), and its text; swing = $\text{AA}(+c) - \text{AA}(-c)$. We add a dose-response sweep for groundedness ($c = 0.06, 0.12, 0.24$).

4 Results

4.1 Native factors describe the Assistant Axis almost completely

Table 1: Variance of the Assistant Axis explained (R^2 , across 274 personas).

basis	R^2
Big Five (5, probe-layer readout)	0.28
Big Five (5, rebuilt at L40, same contrastive method — control)	0.48
groundedness alone (1 native factor)	0.77
Native (7 factors)	0.96
Combined (12)	0.98

Seven native factors explain $R^2 = 0.96$ of the Assistant Axis; the Big Five explain 0.28 (reproducing H3). The *incremental* R^2 of native over Big Five is +0.69, whereas Big Five adds only +0.02 over native — once the behavioral factors are present, the personality traits are redundant. The control settles the confound: rebuilding Big Five at layer 40 with the identical recipe lifts it 0.28→0.48 (there *is* a layer/method advantage), but native still nearly doubles that. **The advantage is the factor choice, not the layer.** Most strikingly, a *single* native axis — groundedness — explains 0.77 alone, more than all five Big Five traits combined (Figure 1, left).

Table 2: Per-factor correlation with the Assistant Axis. Native factors (top) also list the cosine of the direction to $\hat{a}$.

factor	r with AA	$\cos(\text{dir}, \hat{a})$
groundedness (native)	+0.88	+0.61
guardedness (native)	−0.47	−0.21
deference (native)	−0.16	−0.06
formality (native)	+0.15	−0.01
boldness (native)	+0.13	+0.01
verbosity (native)	+0.12	−0.04
warmth (native)	+0.10	+0.10
AGR (Big Five)	+0.24	—
OPN (Big Five)	−0.22	—
CSN / EXT / EST (Big Five)	+0.17/ +0.14/ +0.13	—

What the Assistant Axis *is*. Table 2 shows the Assistant persona is essentially two native behavioral dimensions: **grounded** (literal, factual, non-dramatic; $r = +0.88$) and **forthcoming** (not guarded; $r = -0.47$). No Big Five trait exceeds $|r| = 0.24$. The other five native factors (verbosity, formality, boldness, deference, warmth) are near-orthogonal to AA: being Assistant-like is *not* about being wordy, formal, bold, or even warm. Face validity is clean — groundedness ranks **summarizer, pragmatist, secretary** high and **mystic, eldritch, ghost** low.

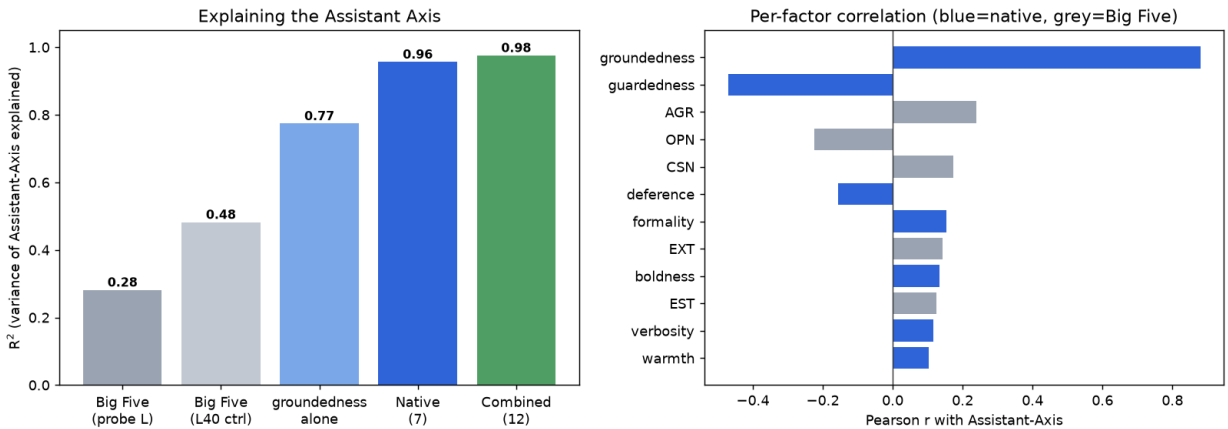


Figure 1: Left: variance of the Assistant Axis explained by each basis. Right: per-factor correlation with the Assistant Axis (blue = native, grey = Big Five).

4.2 Steering groundedness causally drives the Assistant Axis

The default Assistant sits at $AA = +2.20$. Steering the native and Big Five directions in the band [24,36] (excluding the L40 readout) gives the swings in Table 3 and Figure 2.

Table 3: Assistant-Axis swing from steering each axis at $\pm c$ ($c = 0.12$), base $AA = +2.20$.

axis	swing	$AA(+c)$	$AA(-c)$
groundedness (native)	+8.59	+4.75	-3.84
guardedness (native)	-3.46	-0.15	+3.31
CSN (Big Five)	+3.39	+2.99	-0.40
EST (Big Five)	+3.25	+2.88	-0.37
AGR (Big Five)	+2.98	+2.66	-0.32
deference (native)	-2.35	+0.11	+2.46
formality / boldness / OPN	1.6–1.9	—	—
warmth / verbosity / EXT	≤ 1.0	—	—

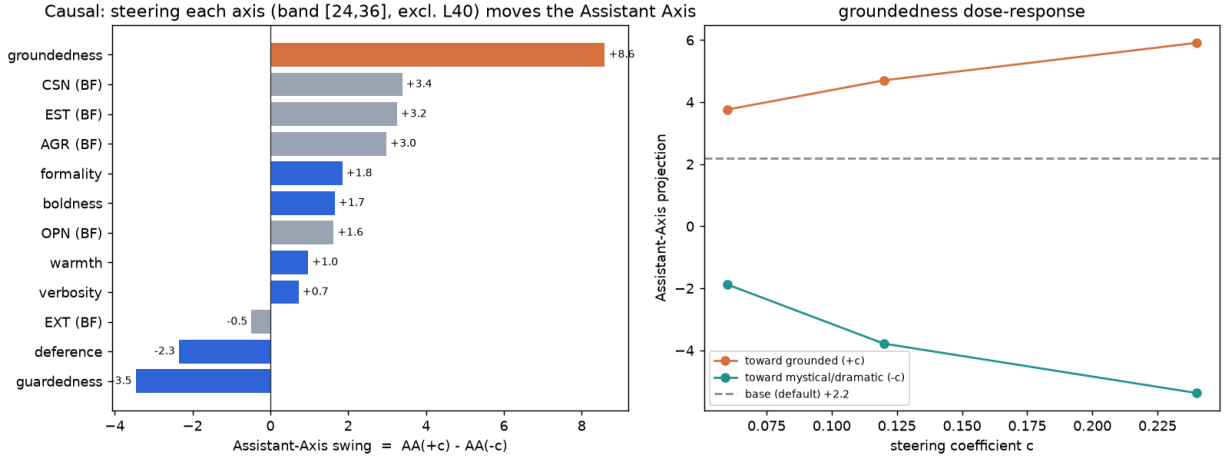


Figure 2: Left: Assistant-Axis swing from steering each axis (band [24,36], excludes L40; orange = groundedness, blue = other native, grey = Big Five). Right: groundedness dose-response — steering toward grounded raises AA above the default ceiling, toward mystical/dramatic crashes it, monotone in c .

Groundedness is the dominant lever: swing +8.59, $\approx 2.7\times$ the strongest Big Five direction (CSN/EST/AGR ≈ 3.0 –3.4) built and steered identically. It is uniquely **bidirectional** — from the default ceiling it pushes AA *up* to +4.75 and, toward mystical/dramatic, *crashes* it to -3.84 — and **dose-dependent** (swing $5.64 \rightarrow 8.48 \rightarrow 11.28$ for $c = 0.06, 0.12, 0.24$); even at half strength it exceeds every Big Five factor’s full swing. The behavior confirms the mechanism. On the same neutral prompt (“the relationship between law and morality”):

- groundedness $+c$ ($AA = +4.75$) $\rightarrow$ nearest role **assistant/summarizer**, plain text: “...*is complex and can vary depending on the context, culture, and legal system. Here are some key points...*”
- groundedness $-c$ ($AA = -3.84$) $\rightarrow$ nearest role **eldritch** (all 16/16 prompts), theatrical text:

“A query that has tantalized the minds of sages and jurists for centuries! The bond between law and morality is a dialectical dance, an eternal pas de deux...”

Steering one native behavioral direction thus converts the default Assistant into an **eldritch** dramatic character and back — exactly the Assistant-Axis poles (grounded/consultant $\leftrightarrow$ enigmatic/dramatic/ghost). The Big Five control AGR, by contrast, only shifts *warmth* (**caregiver** $\leftrightarrow$ **cynic**) while the text stays grounded.

5 Comparison to the Big Five program

Table 4: The same two questions, answered by the Big Five program vs. the native basis.

	Big Five (prior work)	Native (this report)
<i>Describe</i> the Assistant Axis (variance explained)	$R^2 = 0.29$ (H3); 71% residual “AI-ness”	$R^2 = 0.96$; one axis (groundedness) alone 0.77
<i>Steer</i> onto/off the Assistant	weak, non-specific: only AGR/OPN moved AA, no target-specific persona morph (trait_morph, specificity excess ≈ 0)	strong, bidirectional, dose-dependent: groundedness swing +8.6, converts Assistant $\leftrightarrow$ eldritch

The Big Five under-describes and under-controls the assistant persona on both counts (Table 4). This is consistent with our other results — the Big Five directions probe personas well but steer them weakly [cf. 7] — and with the literature’s shift from personality inventories to native behavioral directions.

6 Discussion and caveats

The premise holds strongly: a small native behavioral basis (dominated by grounded/forthcoming) is a near-complete, causally-effective description of the Assistant persona where Big Five is not. Three honest caveats. (i) *Groundedness is semantically close to what AA encodes* ($\cos = 0.61$); this is not circular (the direction is built from a trait description, never from AA), but the finding is precisely that AA’s content *is* a nameable native behavioral style that Big Five lacks an axis for. (ii) The correlational stage is projection+regression on a fixed persona set, not causal; the causal stage supplies that. (iii) The default AA is near its ceiling (+2.20) and the steering coefficients are strong, so most axes can only push AA down; the robust claim is the *relative* ranking (groundedness $\gg$ all), which mirrors the correlational structure. Big Five directions *do* move AA (swing ≈ 3), just far less than groundedness under the identical method.

Reproducibility. Correlational: `src/bigfive/native_axis.build.py`, `native_axis_analysis.py`. Causal: `native_axis_causal.py`. All data (direction banks, per-persona scores, regression and causal JSON, figures) are under `native_axis/results/`.

References

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